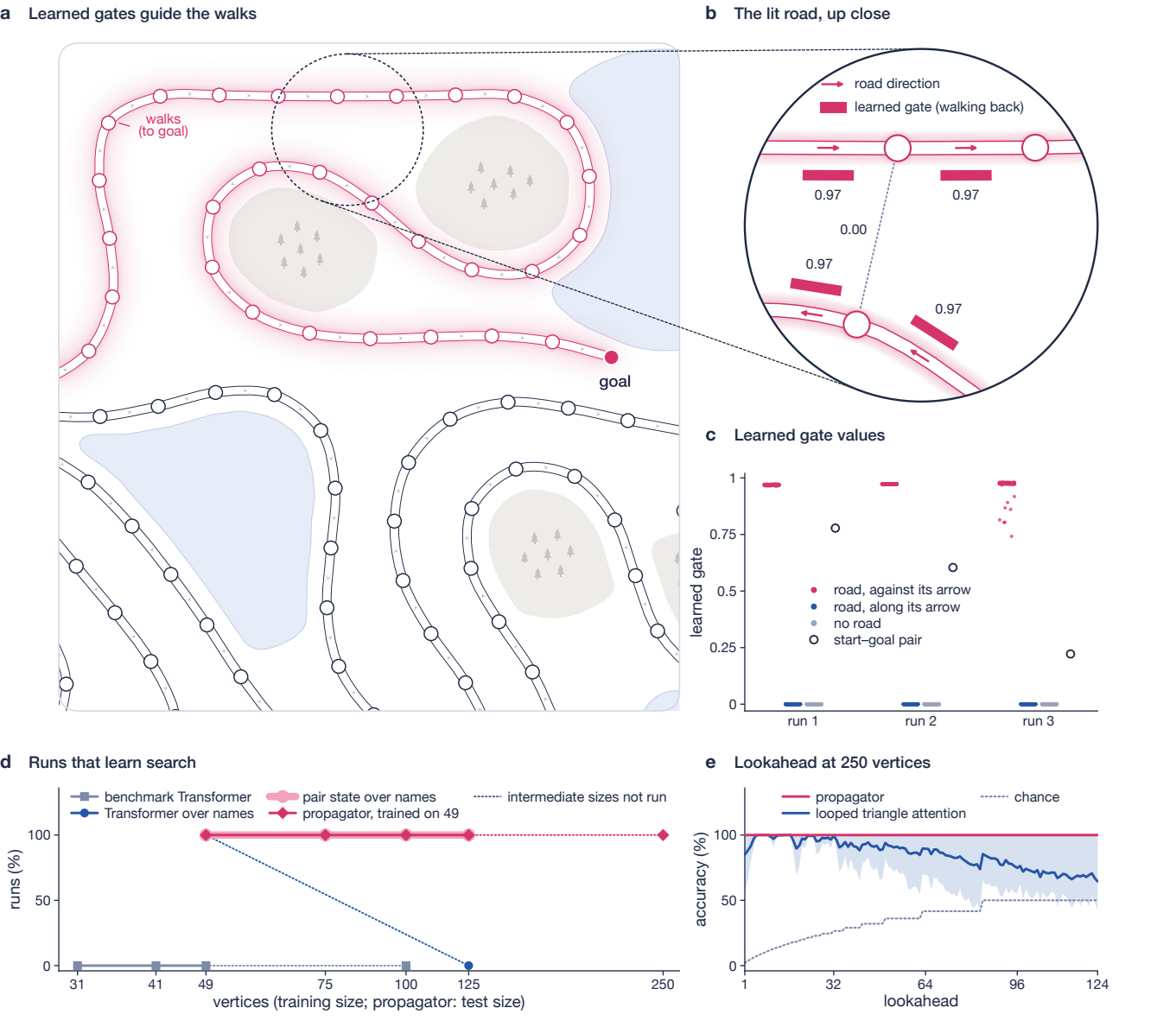

a Learned gates guide the walks
walks
(to goal)
goal
b The lit road, up close
road direction
learned gate (walking back)
0.97
0.97
0.00
0.97
0.97
c Learned gate values
learned gate
1
0.75
0.5
0.25
0
road, against its arrow
road, along its arrow
no road
start–goal pair
run 1
run 2
run 3
d Runs that learn search
benchmark Transformer
Transformer over names
pair state over names
propagator, trained on 49
intermediate sizes not run
runs (%)
100
50
0
31
41
49
75
100
125
250
vertices (training size; propagator: test size)
e Lookahead at 250 vertices
propagator
looped triangle attention
chance
accuracy (%)
100
50
0
1
32
64
96
124
lookahead